```
true
std::ratio_not_equal<std::ratio<1,3>, std::ratio<1,3> >::value ==
false
```

D.6.8. Std::Ratio_Less Class Template

The `std::ratio_less` class template provides a mechanism for comparing two `std::ratio` values at compile time, using rational arithmetic.

Class definition

```
template <class R1, class R2>
class ratio_less:
    public std::integral_constant<bool,see below>
{};
```

Preconditions

`R1` and `R2` must be instantiations of the `std::ratio` class template.

Effects

`std::ratio_less<R1,R2>` derives from `std::integral_constant<bool,` *value* `>`, where *value* is `(R1::num*R2::den) < (R2::num*R1::den)`. Where possible, implementations shall use a method of calculating the result that avoids overflow. If overflow occurs, the program is ill formed.

Examples

```
std::ratio_less<std::ratio<1,3>, std::ratio<2,6> >::value == false
std::ratio_less<std::ratio<1,6>, std::ratio<1,3> >::value == true
std::ratio_less<
     std::ratio<999999999,1000000000>,
     std::ratio<1000000001,1000000000> >::value == true
std::ratio_less<
     std::ratio<1000000001,1000000000>,
     std::ratio<999999999,1000000000> >::value == false
```